

Sedat Demiriz

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Professional Summary

Bioinformatics Programmer with 3 years of experience. Specialty areas include application of reproducible software development methods to solving research problems, identifying associations in genotyping and next-generation sequencing data from general and disease population cohorts, working with complex genomic loci and NGS data

Experience

Computational Biologist - *Sequence Bio Inc.*

Jul '21 – Nov '23

- Lead the development of scalable and reproducible GWAS and ROH analysis pipelines to identify genotype-phenotype associations in company's Newfoundland founder population cohort
- Implemented standalone and pipeline step scripts for genomic data analysis and visualization to identify genes of interest for an international collaboration with pharma company
- Combined internal and public data sources to characterize the distribution of the founder effect across Newfoundland and its relation to source populations in Ireland and Britain

Research Programmer - *BC Genome Sciences Center*

Nov '20 – Mar '21

- Performed QC on inbound AML NGS data using standard NGS quality metrics to ensure clean data input for lab's internal analysis platform
- Streamlined existing QC and visualization scripts and refactored for flexibility of use for future QC efforts
- Addressed graduate student questions about ways of solving AML research questions by showcasing scripting and command line approaches to tackling data transformations

Biosensor R&D Co-op Programmer - *CiBER Lab, SFU*

May – Sep '18

- Automated biosensor voltage response curve data processing and visualization for display and diagnosis of individual sensor instances to identify issues in the fabrication process
- Screened hundreds of sensor instances during development to achieve sample sizes necessary for confidence in manufacturing
- Implemented and presented metrics for determining biosensor instance success based on response pattern to inform lab heads on progress made on the development of the manufacturing process

Genetics Co-op Student - *Istanbul University Genetics Institute*

May – Aug '16

- Carried out patient PCR, DNA and RNA isolation and sample QC procedures for input clinical NGS diagnostics
- Worked routinely in cell culture lab to grow five varieties of tumor cell populations for cancer studies at the lab
- Lead and supervised four other co-op students during their first month with the lab to ensure good integration

Skills

- Reproducible, version-controlled, and containerized data analysis workflows in Python, R, Bash, Github, Apptainer
- Building scalable, modular, and reproducible data analysis pipelines in Snakemake, leveraging HPC resources
- Using field-standard tools for NGS data analysis such as alignment, variant calling, genotyping, and quality control
- Identifying phenotype-genotype associations in clinical data from general population cohorts
- Querying SQL relational databases and unstructured public data sources to derive insights

Education

MSc Human Genetics - *McGill University*

'24 – '26 (expected)

- GPA: 4.0/4.0
- Areas of focus: Human Leukocyte Antigen (HLA) locus, sequencing technologies, software development
- Thesis topic: Reducing NGS data depth biases across HLA locus and other complex genomic loci

BSc Joint Major Mol. Biology and Biochem. & Comp. Sci. - *Simon Fraser University*

'15 – '20

- Computing Science: Data Science, Machine Learning, Databases, Algorithms
- Molecular Biology: Bioinformatics, Human Genomics, Applied Wet Labs
- Statistics: Exploratory Data Analysis, Experimental Design and Analysis

Publications

- **Demiriz, S.** and Taliun, D. (2026). **samsampleX: Distribution-aware downsampling for benchmarking next-generation sequencing data.** *bioRxiv [Pre-print]*

- Gilbert, E., Zurel, H., MacMillan, M. E., **Demiriz, S.**, Mirhendi, S., Merrigan, M., O'Reilly, S., Molloy, A. M., Brody, L. C., Bodmer, W., and et al. (2023). **The Newfoundland and Labrador Mosaic founder population descends from an Irish and British diaspora from 300 years ago.** *Nature Communications Biology*

- Zurel, H., Bhérier, C., Batten, R., MacMillan, M. E., **Demiriz, S.**, Mirhendi, S., Gilbert, E., Cavalleri, G. L., Leach, R. A., Scott, R. E. M., and et al. (2022). **Characterization of the Y chromosome in Newfoundland and Labrador: Evidence of a founder effect.** *European Journal of Human Genetics*

Presentations

Human Genetics Dept. Research Day Poster - *McGill University*

Jun 11, 2026

- Title: samsampleX: pattern-aware NGS data downsampling for depth bias reduction and benchmarking data generation

Master's Thesis Seminar - *McGill University*

Jun 4, 2026

- Title: samsampleX: Distribution-aware downsampling for benchmarking next-generation sequencing data

Canadian Excellence Research Chairs Seminar - *McGill University*

Nov 3, 2025

- Title: Non-uniform downsampling in next-generation sequencing data

Human Genetics Dept. Research Day Poster - *McGill University*

May 27, 2025

- Title: Effects of Whole Exome Genome Sequencing sample multiplexing and whole-genome sequencing depth on HLA genotyping accuracy

Canadian Excellence Research Chairs Seminar - *McGill University*

Mar 3, 2025

- Title: Application of WEGS in Human Leukocyte Antigen (HLA) genotyping (HLA-typing)